

QomL'aqtaqa: A Qom–Spanish Parallel Corpus for Natural Language Processing with Machine Translation Evaluation



Viviana Cotik ^{1,2}, Aleksei Korablev ³, Paola Cúneo ^{4,5}, Pablo Laciana ²

- 1. Universidad de Buenos Aires. FCEN. Departamento de Computación. Argentina.
- 2. CONICET-UBA. Instituto de Ciencias de la Computación (ICC). Buenos Aires, Argentina.
- 3. Minerva University
- 4. CONICET, Argentina
- 5. Universidad de Buenos Aires. FFyL. Instituto de Lingüística, Argentina

1. Motivation



The scarcity of computational resources for Indigenous languages limits the development of technologies that can support their documentation, visibility, and intergenerational transmission.

Qom (also known as Toba):

- ~80,000 speakers across the Argentine Gran Chaco and large urban centers
- Indigenous language of the Guaycuruan family
- Under-resourced language with limited NLP resources.
- No Qom–Spanish parallel corpus suitable for machine translation was previously available.
- Orthographic variation and linguistic divergence from Spanish pose significant MT challenges.

A glimpse of Qom

“Na nqataxaqpi lqataxaqpi na qom da ñapaxaguen na llalec.

‘These are the counsels the Qom give, that they teach to their children.’

Some characteristics:

- **Polysynthetic** and **agglutinative** language with **rich morphology**
- **SVO / VS** word order, differing systematically from Spanish
- **Recently standardized orthography**; variation across available sources.

We contribute with:

- **QomL'aqtaqa**, the first computationally usable Qom–Spanish parallel corpus, accompanied by data statements.
- **A bidirectional Qom↔Spanish MT baseline**
- Corpus and models to be publicly released to support future Qom NLP research

2. The Corpus and Pipeline

33,392

segments

1.47M

Qom tokens

891K

ES tokens

7

sources

Source	Segments
La Biblia	30,449
Arte Verbal	1,831
El Principito	550
Taller Derqui	273
Manual de Salud	212
UDHR	61
Las Aventuras de Copaic	16

Alignments span:

- paragraphs
- sentences
- sentence fragments

Construction Pipeline

Source texts
PDFs - web - printed materials



Extraction
OCR - parsing



Alignment
segments pairing



Linguistic review



QomL'aqtaqa
parallel corpus

3. MT Baseline



References:

- QomL-Base: all sources except the Bible
- QomL-Bible: Bible only

Random train/dev/test split

- **QomL-Base:** 66% / 17% / 17%
- **QomL-Base+Bible:** 80% / 10% / 10%

4. Results (ChrF++)

Direction	Configuration	
	QomL-Base	QomLBase+Bible
ES→QOM	28.8	54.0
QOM→ES	23.1	46.4

Training with Bible data significantly improves MT performance.

Ablation — same non-Bible test set (ChrF++):

Trained on	ES→QOM	QOM→ES
QomL-Base	38.3	30.9
QomL-Bible	33.8	20.3

Training only on Biblical text reduces performance on general-domain content.

5. Conclusions

- QomL'aqtaqa is the first computationally usable Qom–Spanish parallel corpus.
- Bidirectional MT baseline establishes a benchmark for Qom-Spanish translation.
- Domain diversity matters: Bible data improves performance, but general-domain data is essential.
- QomL'aqtaqa enables future research on Qom MT and other NLP tasks.

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